Middle Miocene magnetostratigraphy of the Pebas Formation, Peruvian Amazonia: u-channel records from Tamshiyacu (TAM) and site SC, with palynological and vertebrate (*Purussaurus*) age control

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Abstract

[Placeholder — to be written last.] We present a magnetostratigraphic study of the middle Miocene Pebas Formation in Peruvian Amazonia based on paleomagnetic u-channel records from two sections, Tamshiyacu (TAM; 4.01° S, 73.15° W) and site SC (3.36° S, 71.79° W). Characteristic remanent magnetization (ChRM) directions isolated from despiked, high-coercivity (30 mT to 100 mT) demagnetization identify a well-resolved polarity reversal in each section. Field-based reconstruction of core azimuth recovers geographic declination, so that the reversal is expressed antipodally in both inclination and declination and the virtual geomagnetic poles concentrate toward the geographic poles. Combined with palynological (T-15/T-16) and vertebrate (*Purussaurus*) age control — the latter now anchored by new U-Pb ages from the La Venta Honda Group, where *P. neivensis* occupies chrons C5AAr–C5Ar.2n (∼12.83–13.36 Ma) — the succession is bracketed to ∼12.7–13.8 Ma and correlated to the C5AAr interval of the geomagnetic polarity time scale (∼13.2–13.4 Ma).

1 Geological setting

[Brief setting to complete.] The Pebas Formation (˜Solimões Formation of Brazil) records the middle Miocene Pebas mega-wetland system of western Amazonia — a long-lived lacustrine-to-brackish system with episodic marine influence, deposited in the Andean foreland (Wesselingh & Salo, 2006; Hoorn et al., 2010). Near Iquitos the formation comprises blue-grey lignitic claystones with Pebasian molluscs and marine-influenced palynomorphs, capped by the fluvial Nauta Formation (Rebata et al., 2006). The two sections studied here, TAM and SC, expose this facies along the Amazon River.

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2 Methods

2.1 Sampling and u-channels

Oriented-up (but azimuthally *unoriented*) u-channel samples were collected from the two sections and subdivided into sub-cores separated by foam spacers. Section TAM comprises 12 sub-cores (TAM 1–7, total measured length ∼397 cm); section SC comprises 17 sub-cores (SC 1C–8, total ∼400 cm). Sub-cores were assigned to seven measurement groups (TAM: Groups 1–3; SC: Groups 4–7). Foam-spacer intervals and the outermost ∼5 cm of each sub-core end (u-channel edge effects) were tracked so that they could be excluded from the composite depth scale ("viable positions"); composite depths reported here are foam-corrected sediment depth.

2.2 Paleomagnetic measurements and ChRM isolation

Natural remanent magnetization (NRM) was measured continuously at 1 cm spacing on a 2G Enterprises cryogenic magnetometer at Oregon State University, with in-line alternating-field (AF) demagnetization. Most groups were demagnetized in steps to 100 mT. The characteristic remanent magnetization (ChRM) was isolated by principal component analysis (Kirschvink, 1980) over the high-coercivity interval (30 mT to 100 mT; "300–1000" in instrument units), which removes a soft, steep drilling/coring overprint (Sec. 3.1). Maximum angular deviation (MAD) was recorded for each fit. A u-channel spike (edge-response) removal was applied prior to fitting ("despiked" data). Fits with MAD $> 12^\circ$ were rejected.

Group 1 (TAM 1, 2, 3B) — excluded. This group was measured at only two demagnetization steps (NRM and a single ∼12.5 mT/ 125 Oe step); no high-coercivity series exists, so a ChRM line fit is not possible (the nominal PCA over the 30–100 mT window is degenerate, MAD = 0). The two-step directions are steep and highly scattered (interquartile inclination -52 to $+5^\circ$), dominated by the coring overprint. Group 1 is therefore *excluded* from the polarity record; its polarity is not recoverable without re-measurement with a full AF series. Because Group 1 occupies the base of TAM, the section's uppermost normal interval is not preserved in the ChRM record.

2.3 Polarity determination

The sites lie at $\sim 4^\circ$ S paleolatitude, where a geocentric axial dipole (GAD) field gives a shallow, *negative* inclination for normal polarity; we therefore adopt the convention normal = negative inclination (`POLARITY_SIGN` $= -1$). Inclination is the primary polarity criterion. Absolute declination in the raw core frame is uninformative because the cores are azimuthally unoriented; we recovered it by field-based azimuth reconstruction (Sec. 3.3), which provides a validated second discriminant. ChRM inclinations were smoothed with a 7 cm running median and merged into polarity zones of minimum thickness 12 cm (respecting the ∼5 cm u-channel response function).

2.4 Rock magnetism and magnetic susceptibility

Volume magnetic susceptibility was measured downcore (point sensor, 5 s integration, one iteration per position) for each group. NRM intensity was compared with susceptibility to test whether downcore intensity minima reflect changes in magnetic-mineral concentration (lithology) rather than instrument or field effects.

2.5 X-ray fluorescence (XRF) core scanning

Split-core surfaces were scanned by energy-dispersive micro-XRF (*[instrument model — to confirm]*) with a Mo X-ray source at ∼3 mm resolution. Element intensities (Al, Si, P, S, Cl, K, Ca, Ti, V, Cr, Mn, Fe, . . . , Rb, Sr, Y, Zr, Ba, Pb) are reported as spectral peak areas (semi-quantitative). We use Fe/Ti and related ratios as proxies for magnetic-mineral vs. detrital/clay input, and Ca as a carbonate/mollusc proxy, and compare these against NRM intensity and susceptibility.

2.6 Age model and correlation

Observed polarity zones were pattern-matched to the geomagnetic polarity time scale. We use the astronomically tuned ATNTS2012 (Hilgen et al., 2012), which is identical to GTS2020 (Raffi et al., 2020) in this interval, as the primary scale, and compare against Cande & Kent 1995 (Cande & Kent, 1995) and 1992 (Cande & Kent, 1992) to bound legacy calibration offsets. Independent age control is provided by (i) the western-Amazonian palynological zonation (zones T-15/T-16; Jaramillo et al., 2011, 2017) and (ii) vertebrate biostratigraphy (Sec. 5). All processing was performed with an open Python pipeline (`polarity_synthesis/`, `age_model_ assessment/`; archived with this study).

3 Paleomagnetic results

3.1 Overprint and ChRM

A soft magnetization component removed below 30 mT is steep and of consistent sign, and shallows the direction when stripped (e.g. SC 1C inclination 29° → 20°): it is interpreted as a near-vertical drilling/coring overprint (a mechanical artifact). The high-coercivity ChRM (30 mT to 100 mT) is shallow — per-group median inclinations range −19 to +8° — consistent with the GAD expectation for ∼4° S, and is multi-polarity within the section. Because a single post-depositional overprint cannot produce several reversals, the ChRM is interpreted as a primary (or early) depositional remanence recording the field through deposition.

3.2 Sub-core rotations vs. true reversals

Large declination jumps at sub-core boundaries (e.g. TAM 4→5E ∼182°; TAM 5A→6 ∼140°; TAM 6→7 ∼188°; SC 7B) are accompanied by *no change in inclination sign* and are interpreted as accidental ∼180° or irregular tray rotations of individual sub-cores, not polarity reversals. Genuine reversals require both inclination *and* relative declination to flip and to be near-antipodal; only TAM 3A→4 and SC 4C→4B satisfy this test. SC 4C→4B is the clearest reversal in the dataset: it occurs within a single measurement group (no possible tray change), with excellent MAD (∼4°–5°) on both sides, a ∼157° (near-antipodal) declination offset, and a full inclination flip.

3.3 Core-azimuth reconstruction and VGPs

In the (azimuthally unoriented) core frame, computed VGPs scatter across all latitudes and longitudes rather than clustering near the geographic poles, because the declination is arbitrary and, at these shallow inclinations, VGP latitude is hypersensitive to declination: for a normal, shallow

direction ($I = -8°$) at 4° S, sweeping the unknown declination from 0° to 180° swings the computed VGP latitude from +89° to −83°.

We recovered azimuthal orientation from field measurements of each core's outward (up-facing, in-magnetometer) face relative to the river-bank strike (229°), and rotated the core-frame declination to geographic coordinates ($D = D_{\text{core}} + \text{azimuth} + C$, with one global reference offset $C \approx 35°$ calibrated to a geocentric axial dipole field). Because the outcrop orientation was recorded but the magnetometer tray-loading was not, roughly half the sub-cores additionally required a binary 180° "lab-flip"; this is resolved by inclination (a 180° declination flip does not change inclination) and therefore does not alter the inclination-based polarity.

After reconstruction the directions collapse onto the geocentric-axial-dipole normal and reversed directions (Fig. 1) and the VGPs concentrate strongly toward the poles — the fraction within 30° of a pole rises from 24 % to 69 % (Fig. 2). Individual cores validate well (e.g. TAM 5E to within 5° with no lab-flip; TAM 4 to 9°), confirming the field azimuths carry real signal. The reconstruction thus *confirms* the inclination polarity (TAM 4 reversed; TAM 5–6 normal; SC 4C→4B reversal now antipodal in oriented declination as well as inclination) and provides a validated second discriminant, though because of the residual flip ambiguity at this low paleolatitude inclination remains the primary polarity criterion.

3.4 Polarity records and internal correlation

Foam-corrected composite polarity is:

- **TAM**: (uppermost normal not preserved) → Reversed → Normal (∼99 cm of usable ChRM). The azimuth reconstruction additionally suggests the basal core TAM 7 is *reversed* (Sec. 3.3); if confirmed with better demagnetization this would make the base of TAM Reversed–Normal–Reversed (a second, low reversal), but TAM 7 is currently low-quality and this remains tentative.
- **SC**: Normal → Reversed → Normal (∼262 cm), with the reversal at 106 cm to 149 cm composite depth.

Each section records (at least) one well-resolved reversal, now confirmed in both inclination and reconstructed declination; the most parsimonious interpretation is that TAM's reversal (TAM-4) and SC's reversal (SC-4C→4B) are the same event. SC's Normal–Reversed–Normal thickness ratio (103 : 43 : 88 cm) is best matched by a short reversed chron between two long normals; a pattern-fit favours the C5AAr–C5ABr interval (∼13.2–13.7 Ma). TAM, having lost its uppermost normal, is under-constrained on its own and is consistent with the same tie.

4 Rock-magnetic and XRF results

NRM intensity co-varies with magnetic susceptibility in three of four matched cores (log-intensity vs. susceptibility $r \approx 0.5$–0.75; TAM 3A weaker at 0.23). Because susceptibility scales with magnetic-mineral concentration, downcore intensity minima are interpreted as real sediment-property changes (magnetite-poor, clay- or organic-rich intervals) rather than artifacts. Two intervals show anomalously low or negative susceptibility — SC 7C (susceptibility ∼100× lower than adjacent cores) and SC 2 (diamagnetic) — and carry correspondingly unreliable directions; these are masked in the polarity interpretation.

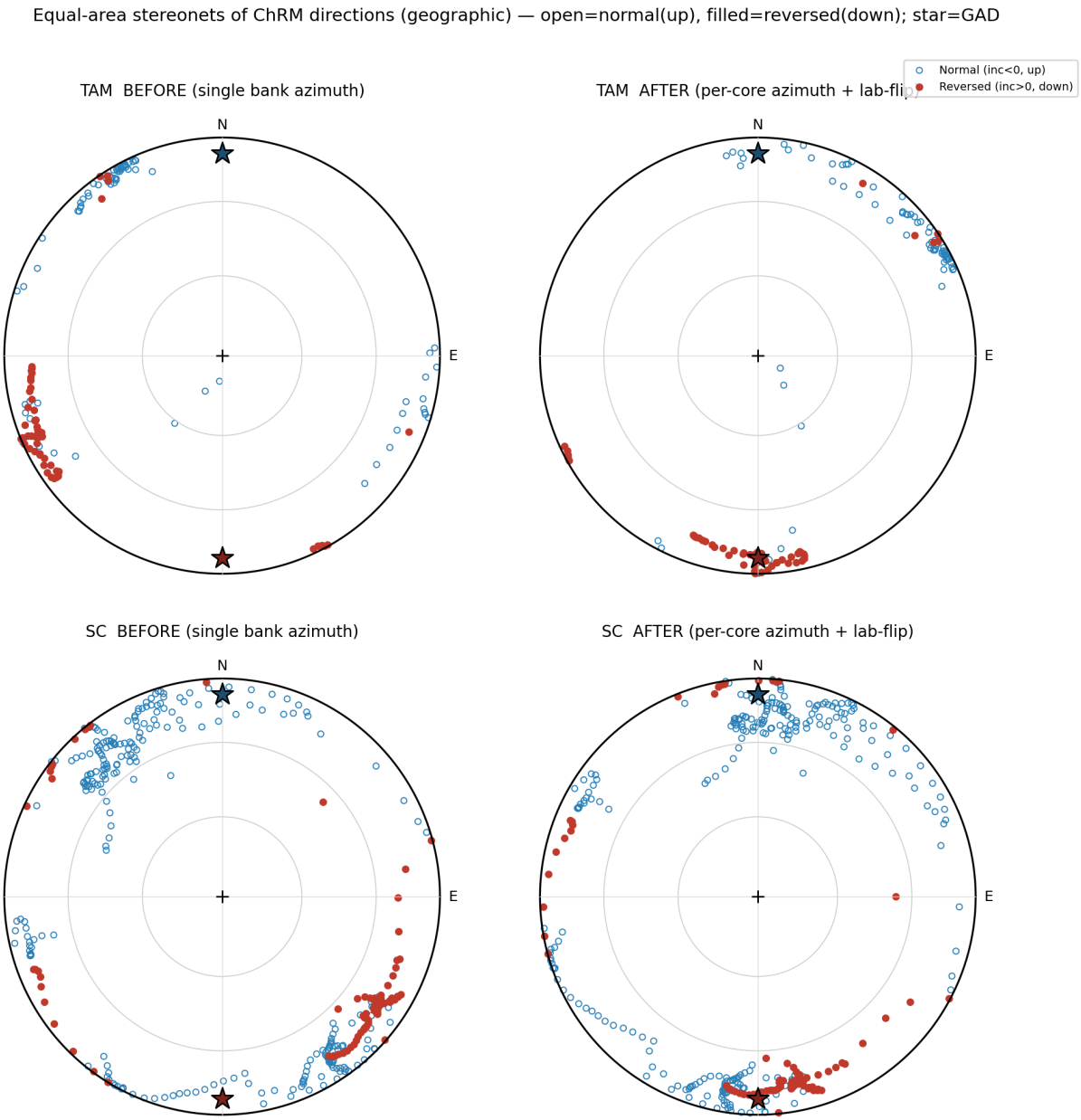


Figure 1: Equal-area stereonets of ChRM directions (geographic frame), before (single bank azimuth) and after per-core azimuth reconstruction (with the binary lab-flip resolved). Open = normal (upward, $I < 0$); filled = reversed (downward, $I > 0$); stars = expected GAD normal/reversed directions. After reconstruction the directions collapse onto the N–S (declination 0/180°) axis at the GAD directions.

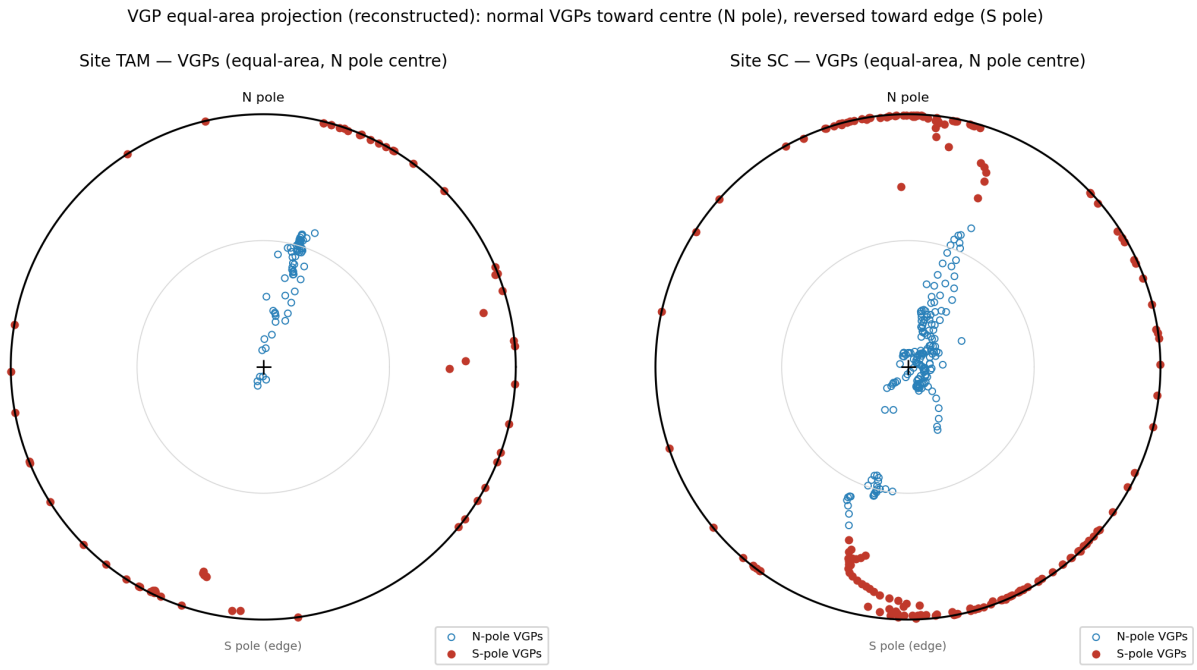


Figure 2: VGP equal-area projection (N pole at centre) of the reconstructed directions. Normal-polarity VGPs (open) fall toward the centre (north pole) and reversed VGPs (filled) toward the edge (south pole), for both sites.

The 27 XRF scans (1906 validity-filtered points) were depth-registered to the paleomagnetic record per group (XRF position/10 = measured u-channel position). Contrary to the initial expectation that magnetic-mineral (Fe) content would predict remanence intensity, **total Fe is only weakly and inconsistently related to NRM intensity** (per-group Pearson r on log-log -0.10 to $+0.45$; pooled, within-group-standardized $r \approx +0.16$; Fe/Ti $r \approx +0.08$). This is not a registration artifact: a depth-lag scan peaks at zero offset and a coarse (5 cm-binned) comparison gives the same weak result ($r \approx +0.18$).

Instead, **Fe co-varies with S and Ca**: log Fe vs. log S reaches $r = +0.5$–0.7 (pooled $+0.39$) and log Fe vs. log Ca reaches $r = +0.7$–0.9 (pooled $+0.56$). We interpret this to mean that in this organic-rich, reducing, lignitic facies most iron resides in *authigenic* phases — pyrite (Fe–S) and carbonate/shell-associated or Fe-carbonate intervals (Fe–Ca) — rather than in detrital magnetite. Total Fe measured by XRF is therefore a poor proxy for the remanence carrier here, which is why it fails to track NRM intensity even though *susceptibility* does ($r \approx 0.5$–0.75; above). The two groups with the strongest Fe–intensity coupling (TAM 5B–5A–6–7, $r \approx 0.45$; SC 3B–3A–4, $r \approx 0.27$) are also those with the strongest Fe–S/Fe–Ca signal, i.e. where a larger detrital component is preserved.

The implication is that reductive diagenesis (pyritisation) exerted a first-order control on the magnetic mineralogy: it both sequestered iron into pyrite and diluted or dissolved the detrital magnetite that carries the remanence, producing the low-intensity, low-susceptibility intervals. The lithological control on remanence intensity is thus real, but it is diagnosed by susceptibility, not by XRF Fe; XRF instead resolves the diagenetic (pyrite) and carbonate/detrital architecture of the section. Downcore profiles and cross-plots are given below (Figs 4, 5).

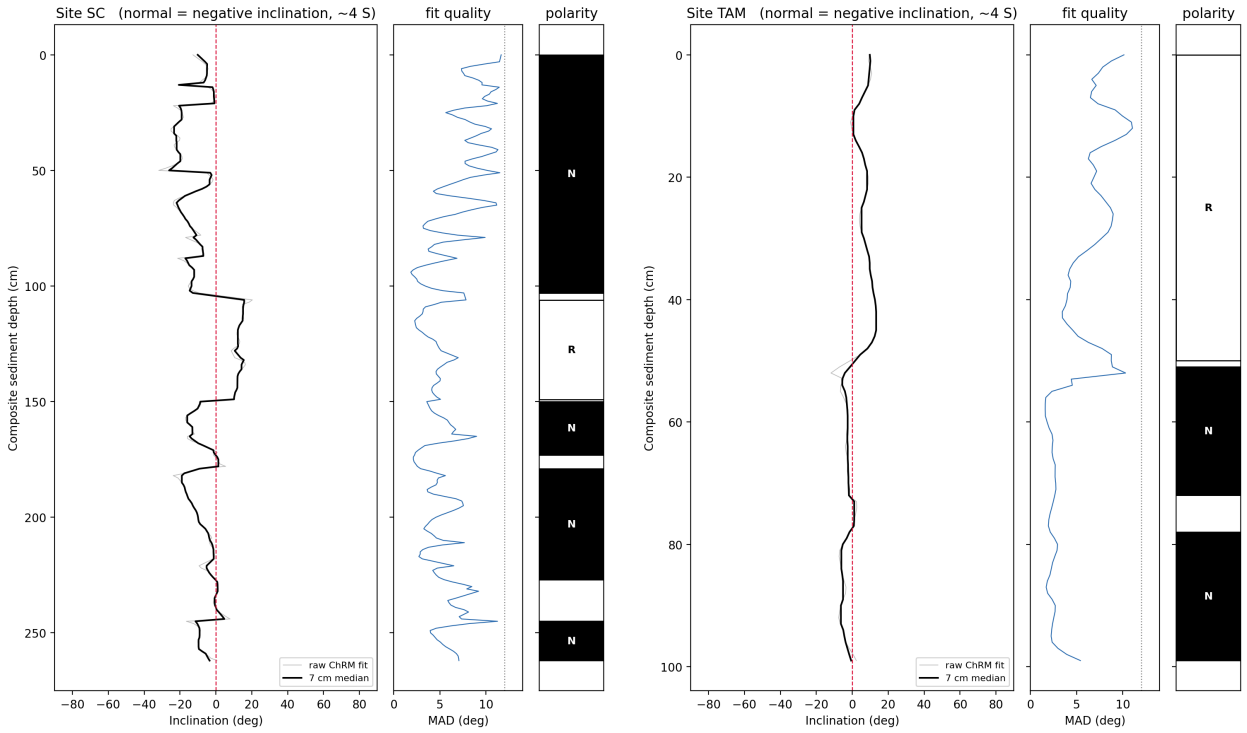


Figure 3: Corrected ChRM inclination (raw grey; 7 cm running median black), MAD, and the resulting polarity column against composite sediment depth for sites SC (left) and TAM (right). Normal polarity = negative inclination at ~4° S. SC records Normal–Reversed–Normal with the reversal at 106 cm to 149 cm; TAM records Reversed–Normal (its uppermost normal was not preserved; Group 1 excluded).

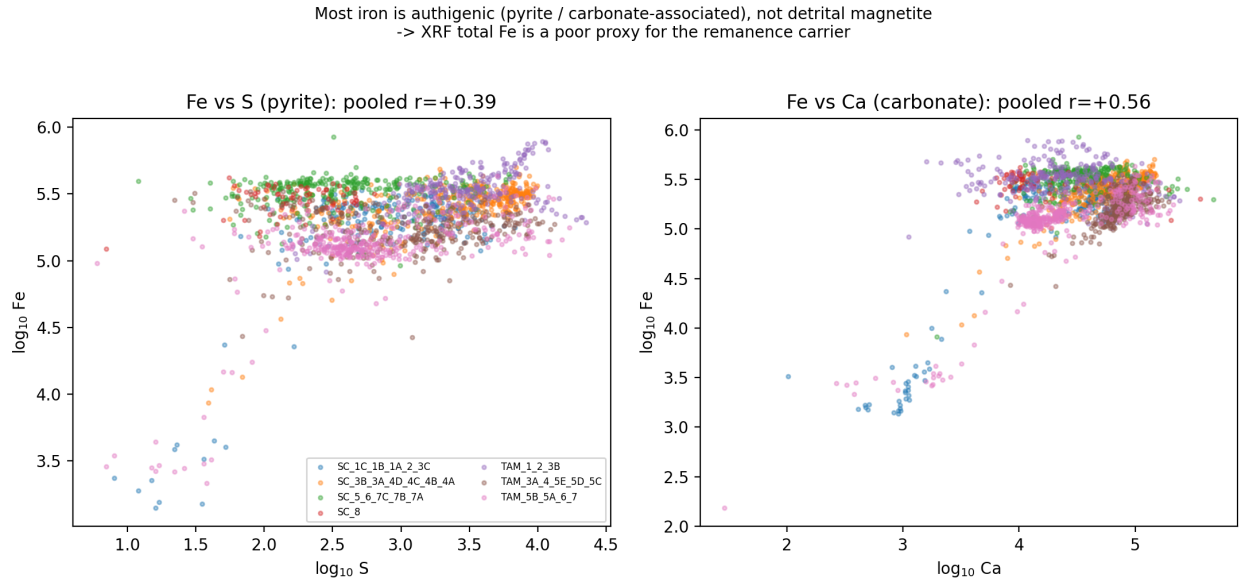


Figure 4: XRF total Fe co-varies with S (pyrite; pooled $r = +0.39$) and Ca (carbonate; pooled $r = +0.56$), indicating that most iron is authigenic rather than detrital magnetite. This is why total Fe does not predict the magnetite-controlled remanence intensity.

Table 1: Reversal reliability summary.

Boundary	Inc. flip	Rel. decl. flip / antipodal	Verdict
SC 4C→4B	yes	yes (∼157°)	**true reversal (high)**
TAM 3A→4	yes (gradual)	partial	true transition (moderate)
TAM 4→5E	no	∼182°	rotation artifact
TAM 5A→6	no	∼140°	rotation artifact
TAM 6→7	possible	∼188°*	possible N→R reversal (tentative)
SC 7B	no	∼180°	rotation artifact

*After azimuth reconstruction TAM 7 fits a reversed direction for any orientation consistent with its neighbours, so the TAM 6/7 declination jump may be a genuine N→R reversal rather than only a tray rotation; TAM 7 is low-quality, so this is tentative.

4.1 Elemental stratigraphy

Downcore, the elements most diagnostic of the depositional environment vary by one to two orders of magnitude and co-vary systematically (Fig. 6). Sulphur (pyrite) ranges over ∼two orders of magnitude (10th–90th percentile ∼110–6000 peak-area units) and is highest in discrete, decimetre-scale intervals; Fe and Ca track those same intervals (pooled log–log r: Fe–S = +0.39, Fe–Ca = +0.56, S–Ca = +0.57), and Ca in turn co-varies with Sr (r = +0.62), the expected signature of aragonitic mollusc carbonate. Fe/Ti (authigenic-Fe enrichment above the detrital baseline) rises in the same pyrite-and-carbonate-rich intervals, whereas K/Ti (a clay/detrital proxy) is comparatively flat, indicating the elemental variability is driven by authigenic mineral formation rather than changing detrital provenance. Manganese, mobile under reducing pore-waters, is low in the most pyritic intervals.

5 Fossil age control: *Purussaurus*

A fossil of the giant caiman *Purussaurus* was recovered at TAM approximately 0.5 m above the base of the measured section. *Purussaurus* is a middle-to-late Miocene genus whose oldest records (∼13 Ma; *P. neivensis*) occur at La Venta (Colombia) and in the Pebas Formation near Iquitos, where *P. neivensis* is the apex predator of a ∼13 Ma crocodylian assemblage dated by palynology and mollusc/ostracod zonation (Salas-Gismondi et al., 2015; Pujos & Salas-Gismondi, 2020). The genus therefore imposes a maximum-age (older) bound of ∼13.8 Ma on the fossil horizon — the complement of the palynological T-15/T-16 minimum-age bound of 12.7 Ma. Together these independent constraints bracket the section to ∼12.7–13.8 Ma (and, using the ∼13 Ma Iquitos-Pebas assemblage age, to ∼12.7–13.0 Ma), consistent with the magnetostratigraphic correlation to C5AAr–C5ABr (Fig. 7). The independent, U-Pb-based refinement of this ceiling from La Venta is developed in Section 5.1.

Caveats: (i) genus-level identification alone permits only the broad mid–late Miocene (∼13–6.8 Ma) range; the tight bound requires species identification, as only *P. neivensis* is diagnostically middle Miocene (a late Miocene species — *P. mirandai*, *P. brasiliensis* — would falsify the ∼13 Ma reading). (ii) Because *Purussaurus* in Amazonia is partly calibrated by the Pebas itself, the independent leg of the argument is La Venta, where *P. neivensis* is now dated by U-Pb zircon geochronology (Section 5.1; Mora-Rojas et al., 2023), not only mammal biostratigraphy. (iii) The fossil lies in the (excluded) Group 1 interval, so it presently anchors the section as a whole rather

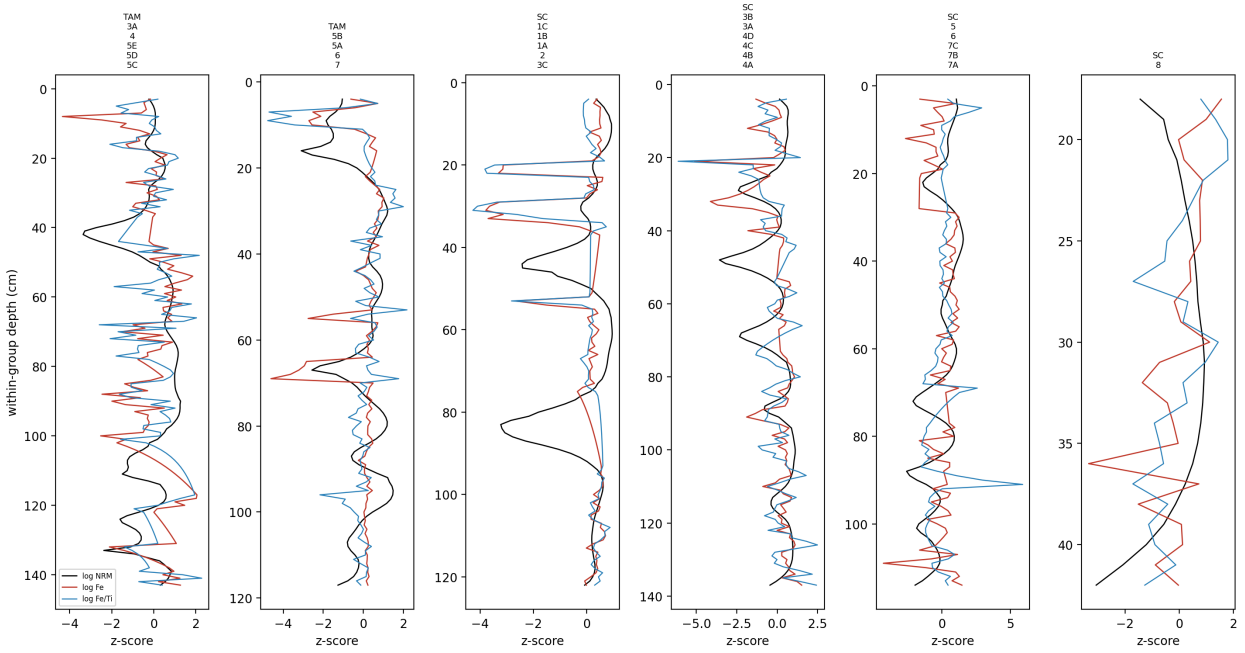


Figure 5: Downhole (within-group z-scored) NRM intensity vs. Fe and Fe/Ti. The lack of consistent coincidence between intensity and Fe minima confirms that XRF Fe is a poor proxy for the remanence carrier; susceptibility (not shown) remains the reliable magnetic-concentration proxy.

than a specific polarity zone.

The Pebas facies (blue-grey lignitic clays, Pebasian fauna) together with this Pebas-fauna vertebrate exclude a younger fluvial-terrace or Nauta-Formation origin for the section; the terrace, Nauta, and Madre-de-Diós units of the region overlie, and are younger than, the sediments studied here.

5.1 Regional correlation to La Venta (Honda Group)

The strongest *independent* calibration of *Purussaurus neivensis* comes not from the Pebas (which would be circular) but from La Venta in the Tatacoa Desert, Colombia, whose Honda Group has recently been re-dated with new U-Pb zircon geochronology and a Bayesian age–depth model (Mora-Rojas et al., 2023; Carrillo et al., 2023). In that framework the La Victoria Formation spans <16 to 12.58 Ma and the Villavieja Formation 12.58 to 10.52 Ma (Table 2), and the magnetostratigraphy is reinterpreted as chrons C5ABn (base) through C5An.1n (top) on the ATNTS2012/GTS2020 scale. Critically, the fossil-rich "Monkey Beds" that yield *P. neivensis* are assigned to chrons **C5AAr–C5Ar.2n (∼12.83–13.36 Ma)**. Independently, Carrillo et al. (2023) note that Laventan-equivalent beds are now recognised at *Tamshiyacu* itself (near Iquitos), sharing taxa with La Venta.

This lets us tie the Peru record to a regional, radiometrically calibrated reversal barcode rather than to the abstract GPTS alone (Fig. 8). The preferred correlation places the Peru reversal in **C5AAr** (∼13.2–13.4 Ma): it satisfies (i) the SC polarity-pattern fit, (ii) the palynological floor, (iii) the *Purussaurus* ceiling, and (iv) the U-Pb-dated La Venta *P. neivensis* interval, which all overlap at ∼13.2–13.4 Ma. This converts the earlier tentative ∼13 Ma tie into one anchored on zircon dates.

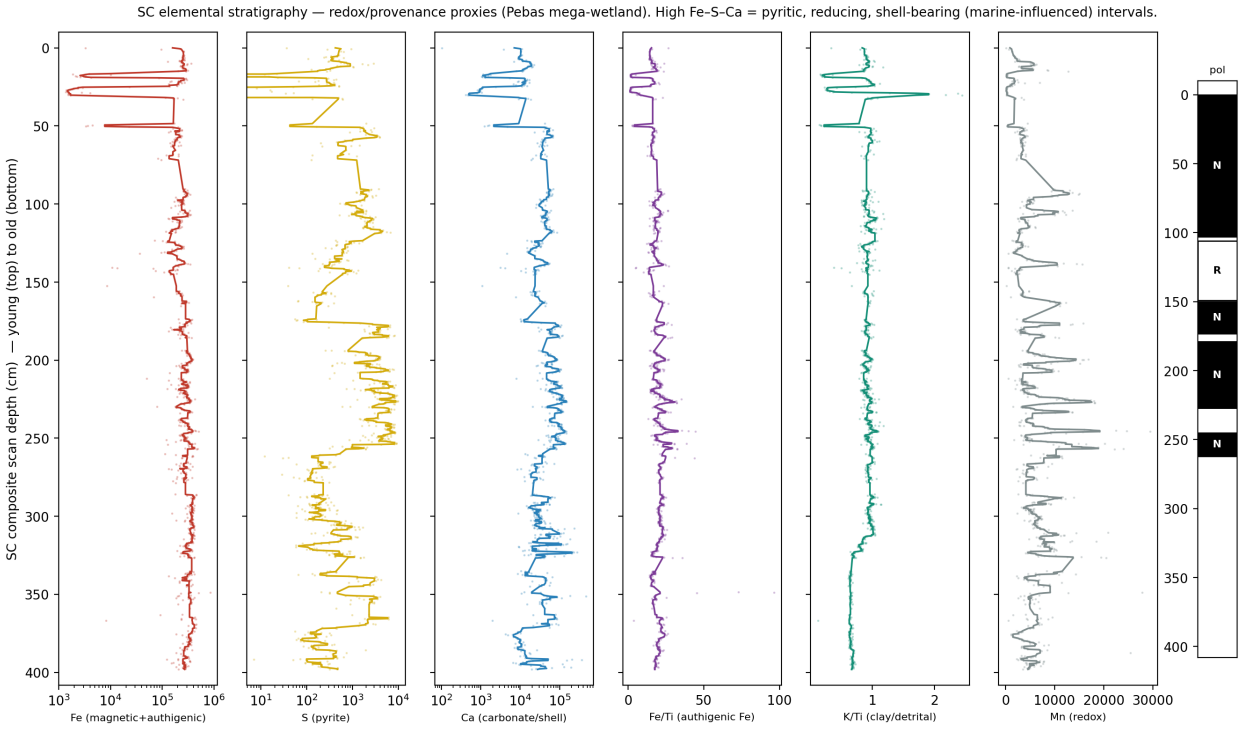


Figure 6: Elemental stratigraphy of the SC composite (valid XRF points; running median). Sulphur (pyrite), Fe and Ca (carbonate/shell, with Sr) are enriched together in discrete reducing intervals; Fe/Ti (authigenic Fe) tracks them while K/Ti (detrital) stays flat and Mn (redox) is depressed. Polarity column at right. Depth is approximate composite scan depth (young at top).

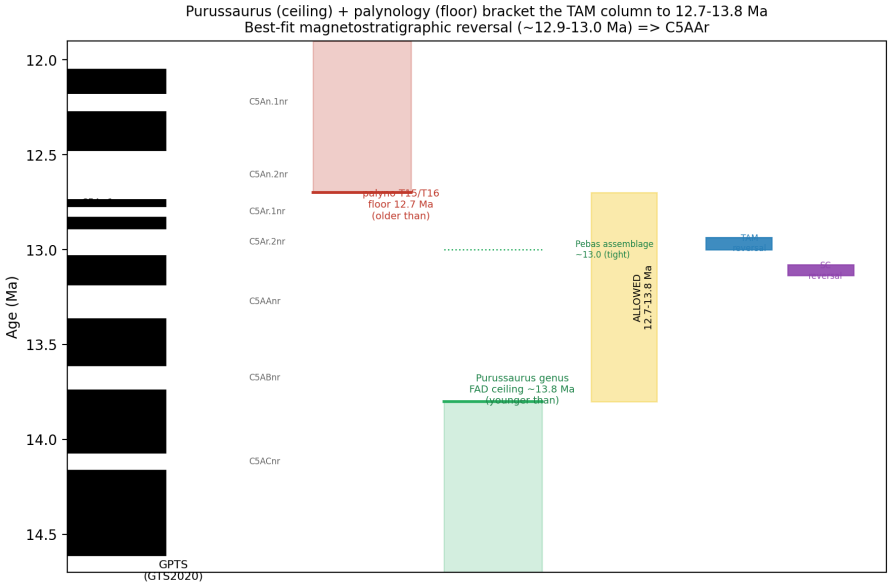


Figure 7: The *Purussaurus* genus ceiling (<∼13.8 Ma) and the palynological T-15/T-16 floor (>12.7 Ma) bracket the section to ∼12.7–13.8 Ma; the tighter Iquitos-Pebas assemblage age (∼13 Ma) and La Venta *P. neivensis* interval narrow it further. The best-fit magnetostratigraphic reversal falls at the young edge of this bracket.

Table 2: U-Pb zircon ages for the Honda Group (La Venta), from Mora-Rojas et al. (2023). Median and 95% CI from the Bayesian age model; YSP = youngest statistical population.

Sample	Strat. (m)	Age (Ma, 95% CI)	Type
TVV-01	932	11.5 (10.52–12.09)	detrital MDA
SEL-02	503	12.3 (11.79–12.74)	tuff, YSP 12.8±0.1
TVV-04	474	12.3 (11.80–12.77)	tuff, YSP 12.8±0.1
44011	218.8	13.1 (12.69–13.47)	tuff, YSP 13.9±0.3
44017	63.5	16.0 (15.94–16.00)	tuff, YSP 17.6±0.4

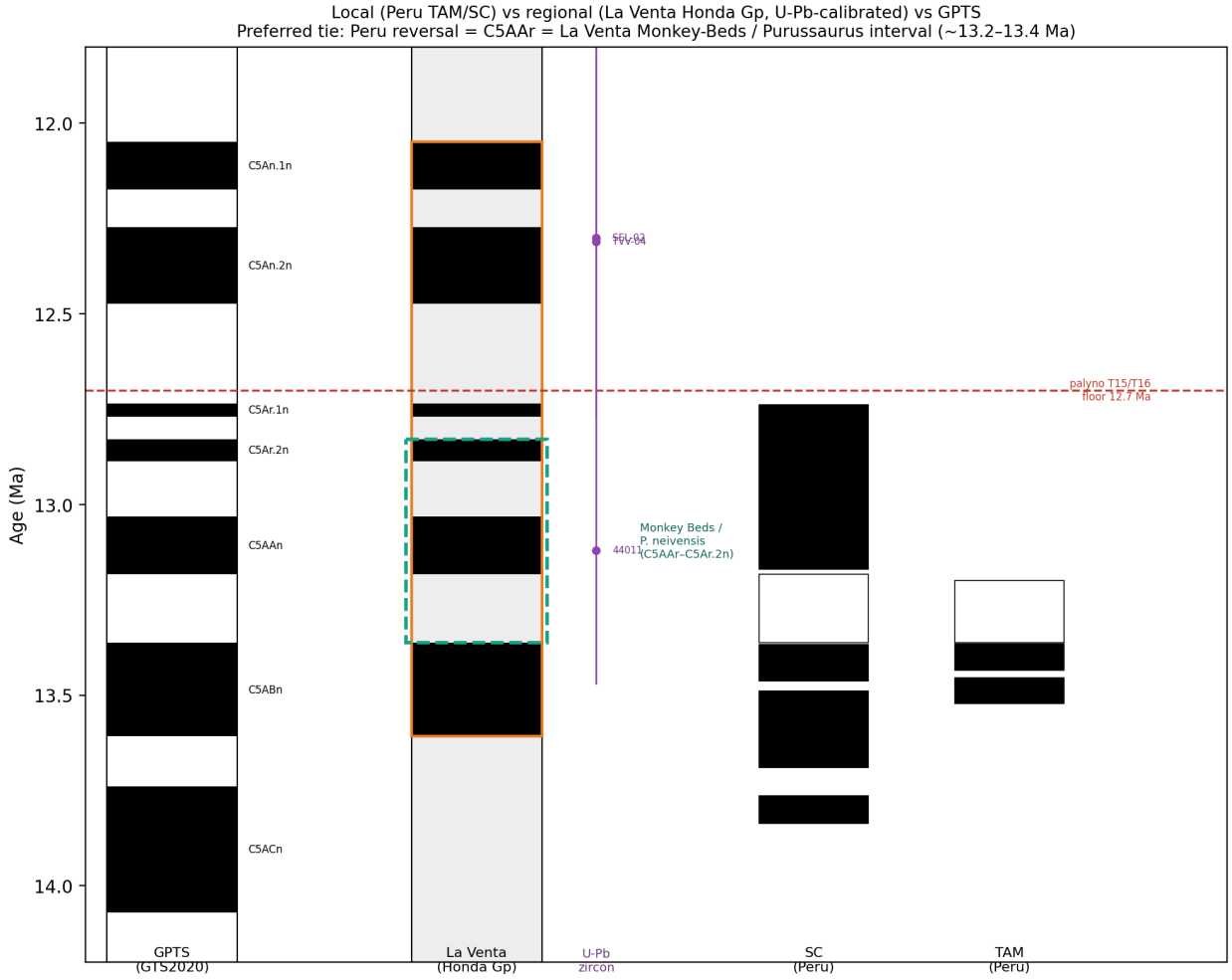


Figure 8: Local (Peru TAM, SC) vs. regional (La Venta Honda Group, U-Pb-calibrated) vs. GPTS (GTS2020) polarity barcodes on a common age axis. Black = normal. The La Venta column shows the occupied interval (C5ABn–C5An.1n, orange), the *Purussaurus*-bearing Monkey Beds (C5AAr–C5Ar.2n, green dashed), and U-Pb sample ages (purple). The Peru reversal aligns with C5AAr and the La Venta Monkey-Beds interval; the palynological floor (12.7 Ma, red) bounds it from below.

6 Discussion

6.1 Age of the Pebas Formation at Tamshiyacu

Four independent lines of evidence converge on a middle Miocene age of ∼13.2 Ma to 13.4 Ma (chron C5AAr) for the studied succession. (i) The polarity record: each site preserves a single robust reversal, now expressed antipodally in both inclination and reconstructed declination, and SC's Normal–Reversed–Normal thickness ratio favours a short reversed chron between two long normals (C5AAr–C5ABr). (ii) Palynology places the section in zone T-15 / base T-16, a firm minimum-age (younger-than) bound of 12.7 Ma. (iii) The giant caiman *Purussaurus* supplies the complementary maximum-age (older-than) bound: the genus does not appear before ∼13.8 Ma. (iv) Most decisively, new U-Pb geochronology from La Venta (Mora-Rojas et al., 2023) dates *P. neivensis* (the same species reported from the Iquitos Pebas) to chrons C5AAr–C5Ar.2n (∼12.83 Ma to 13.36 Ma), an anchor independent of the Pebas itself and therefore free of the circularity that weakens a Pebas-internal argument. These four constraints overlap only around C5AAr, and the independent recognition of Laventan-equivalent beds at Tamshiyacu (Carrillo et al., 2023) corroborates the correlation. The middle-Miocene calibration is robust to the choice of GTS2020/ATNTS2012 vs. CK95 (a 50 ka to 100 ka shift), which is smaller than the current tie uncertainty.

6.2 Magnetic mineralogy and diagenetic overprinting

The ChRM is a shallow, dual-polarity, primary detrital remanence, but its intensity is strongly modulated by diagenesis rather than by the primary depositional flux of magnetic minerals. NRM intensity tracks susceptibility ($r \approx 0.5$–0.75) — both measuring magnetite concentration — yet is essentially uncorrelated with XRF total Fe, because in this facies most iron is locked in *authigenic* phases (pyrite and carbonate; Sec. 4.1) rather than in detrital magnetite. The most pyritic intervals are also the lowest in remanence intensity and susceptibility, implying that bacterial sulphate reduction consumed reactive iron and partly dissolved detrital (titano)magnetite, driving the low-intensity zones. This is the expected behaviour of an organic-rich, reducing sediment and it explains why polarity in those intervals is noise-prone and best read from the surviving high-coercivity ChRM. It also means XRF Fe cannot substitute for rock-magnetic measurements as a magnetic-concentration proxy here — a transferable caution for magnetostratigraphy in reducing tropical wetland deposits.

6.3 A freshwater mega-wetland with episodic marine influence

The elemental stratigraphy is best read within the framework of the Pebas mega-wetland (Hoorn et al., 2010; Wesselingh & Salo, 2006): a vast, long-lived, low-gradient system of shallow lakes, swamps and floodplains that covered much of western Amazonia through the middle Miocene — in scale and dynamics the closest modern analogue is a greatly enlarged Everglades-type wetland rather than a coastal or deltaic setting. The system was *dominantly freshwater to oligohaline*, as shown by its hyperdiverse endemic mollusc fauna (cochliopid gastropods, pachydontine and corbulid bivalves), whose diversity and endemism themselves require a persistent, stable, largely freshwater water body (Wesselingh & Salo, 2006). Marine influence was real but *episodic and subordinate* — short brackish ingressions punctuating a freshwater background rather than a standing marine connection.

The XRF cycles record exactly this behaviour. Against a reducing, organic-rich freshwater sub-

strate — the natural state of a peat/marl mega-wetland — discrete intervals enriched in S, Fe and Ca (with Sr) mark the episodic events. Authigenic pyrite (Fe–S) forms where bacterial sulphate reduction operates below the sediment–water interface, and in a freshwater system its abundance is limited chiefly by the scarce supply of dissolved *sulphate*; the strong, punctuated pyrite enrichments therefore most plausibly track brief pulses of enhanced sulphate availability, i.e. the brackish ingressions independently inferred from molluscan and geochemical proxies (Wesselingh & Salo, 2006). Slow net accumulation (below) reinforces this: prolonged residence below the redoxcline in a sediment-starved wetland favours pyritisation even where sulphate is only intermittently supplied. The co-enrichment of Ca and Sr in the same beds — aragonitic Pebasian mollusc coquinas — ties shell beds and pyritic muds to the same low-energy, reducing episodes, while flatter K/Ti and higher Mn between them indicate more oxygenated, detritally-diluted freshwater intervals. The section thus does not record a static environment, nor a marine-dominated one, but a persistent freshwater mega-wetland briefly and repeatedly perturbed by marginal-marine incursions during peak inundation.

6.4 Sedimentation rate, paleoclimate, and the road to the modern Amazon

The polarity–GPTS age model implies remarkably consistent accumulation rates at the two sites — ∼7.9 m/Myr at SC (262 cm in ∼333 kyr) and ∼7.8 m/Myr at TAM (99 cm in ∼127 kyr), or roughly 16–23 m/Myr once decompacted. Far from anomalous, these low rates are what a mega-wetland *predicts*: a flat, poorly-drained, sediment-starved system far from its Andean source accumulates slowly, and the figures are broadly consistent with regional estimates of tens of m/Myr for the Pebas/Solimões over its several-million-year life (Hoorn et al., 2010). (A coastal or deltaic setting would accumulate an order of magnitude faster; the observed rate is diagnostic of the wetland interpretation, not at odds with it.) The internal consistency between the two sites, and the fact that forcing the SC reversed zone to represent an entire reversed chron would require an implausibly slow ∼2.4 m/Myr, together indicate a condensed but coherent record with minor diastems rather than steady, complete deposition — again the expected signature of a low-accommodation wetland.

Climatically, the ∼13 Ma interval sits immediately after the Middle Miocene Climatic Transition (MMCT, ∼14.2–13.8 Ma), the global cooling that ended the Middle Miocene Climatic Optimum; in western Amazonia this coincided with the maximum extent of the mega-wetland, sustained by Andean-foreland subsidence and a very low regional gradient. Our record captures the *peak-wetland* phase of a longer landscape trajectory: the Pebas system persisted through the middle Miocene before the late-Miocene reorganisation of Amazonian drainage, when the establishment of the modern transcontinental Amazon River and the growth of its deep-sea fan (∼11–9.4 Ma; Figueiredo et al., 2009; Hoorn et al., 2010, 2017) drained and replaced the wetland with the fluvial system seen today. Placed on the magnetostratigraphic age model, the Tamshiyacu succession provides a polarity-calibrated tie-point within the Pan-Amazonian framework (La Venta, Contamana, the Marañón and Solimões basins) near the apex of the Pebas mega-wetland, against which the timing of individual incursions and the onset of drainage reorganisation can be tested.

6.5 Limitations and next steps

The GPTS tie rests on a single well-resolved reversal per site; recovering a second reversal in one continuous section (e.g. confirming the tentative TAM 7 reversal with fuller demagnetisation, or re-measuring the excluded TAM Group 1) would make the correlation unique rather than best-fit. Species-level confirmation of the Tamshiyacu *Purussaurus* would convert the vertebrate constraint

from a genus-level ceiling into a tight (∼12.8 Ma to 13.4 Ma) tie. The azimuth reconstruction would tighten with a physically fixed reference offset (from confirming the x- vs y-face convention) and a re-check of the anomalous TAM 6 field azimuth. Finally, the XRF–environment interpretation is currently qualitative and depth-registered only approximately; quantitative sulphate/marine-influence inference would benefit from paired $^{87}Sr/^{86}Sr$ or S-isotope measurements on the pyritic and shell-bearing intervals identified here.

7 Data and code availability

Processing scripts and derived data (polarity zones, age-model scenarios, multi-timescale chron tables, fossil-bracket analysis) are archived in the project repository (`polarity_synthesis/`, `age_model_assessment/`) and in `Uchannel_notebook.ipynb`.

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